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Pharmacology: Sedative-Hypnotics

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Session Objectives

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- **Identify sedative-hypnotic drugs.**
- **Describe MOA, use, and adverse effects of barbiturates.**
- **Describe MOA and adverse effects, and specific uses of benzodiazepines (BDZ).**
- **Describe the differences between barbs, BDZs, and ethanol in MOA and adverse effects. Include management of overdose, withdrawal, and abstinence.**
- **Describe non-sedating anxiolytic drugs.**
- **Describe drugs used to induce sleep.**
- **Describe drugs for narcolepsy.**

Source: Course Syllabus

Sedative-Hypnotic Drugs

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- Barbiturates
- Benzodiazepines
- Ethanol
- Non-benzodiazepines (zolpidem, zaleplon, eszopiclone)
- 1st generation antihistamines
- Newer drugs (melatonin agonists, orexin antagonists)

Degree of sedation-hypnosis depends on dose

Sedative-Hypnotic Drugs

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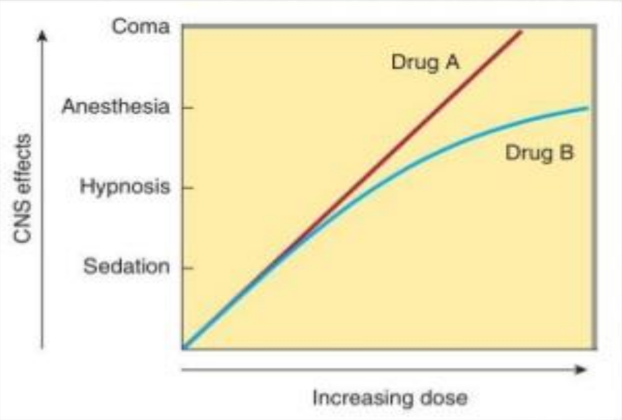
Dose-response curves for two hypothetical sedative-hypnotics



- Drug A:
  - An increase in dose higher than that needed for hypnosis may lead to a state of general anesthesia.
  - With higher doses, the drug will depress the respiratory and vasomotor centers which leads to coma.
  - Drug A is an example of alcohol and barbiturates.
- Drug B:
  - needs greater doses to achieve CNS depression.
  - Drug B is an example of benzodiazepines and newer hypnotics.

Source: Sedative-Hypnotic Drugs, Basic & Clinical Pharmacology, 14e
Citation: Katzung BG. Basic & Clinical Pharmacology, 14e; 2017 Available at: http://accessmedicine.mhmedical.com/content.aspx?bookid=2249&sectionid=175218773 Accessed: December 08, 2022

Barbiturates

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- Pentobarbital, Secobarbital, Phenobarbital, Amobarbital, Thiopental
- Lipid soluble drugs (thiopental) are redistributed from brain.
- Elderly and those with hepatic disease have increase in $t_{1/2}$.
- CYP inducers (3A4)
- Exacerbation of porphyria
- Use: anxiety and insomnia (fallen out of use), seizures (phenobarbital), induction of anesthesia (thiopental).

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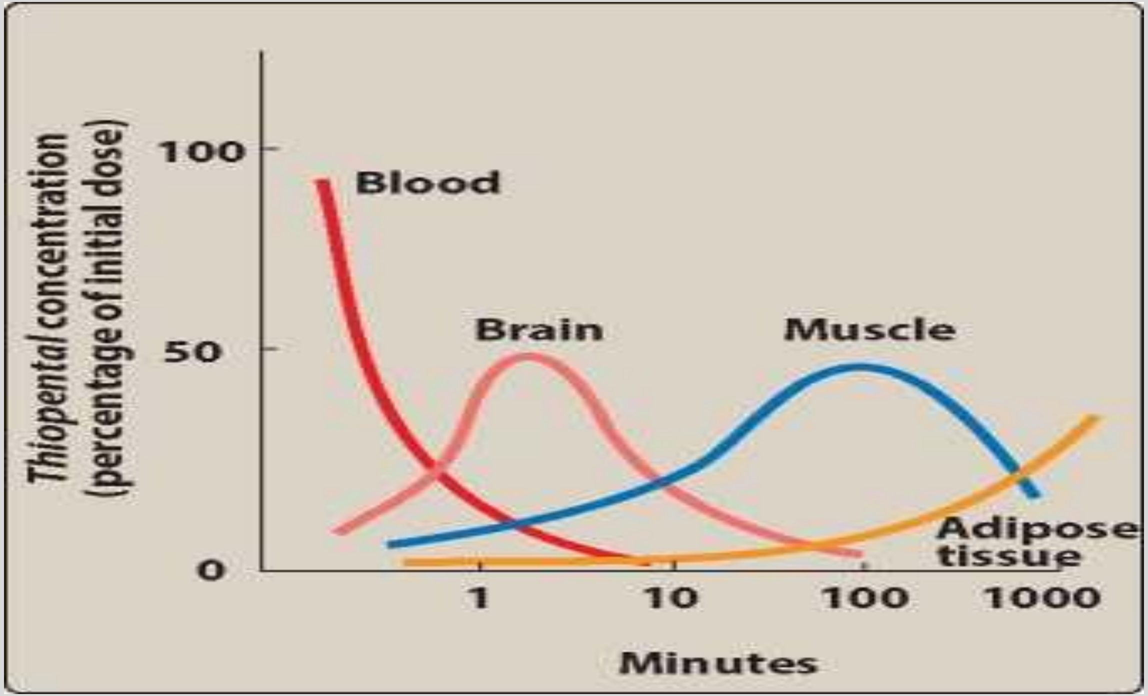

Lippincott Illustrated Reviews: Pharmacology 6th edition (Lippincott Illustrated Reviews Series) 6th Edition

Barbiturates

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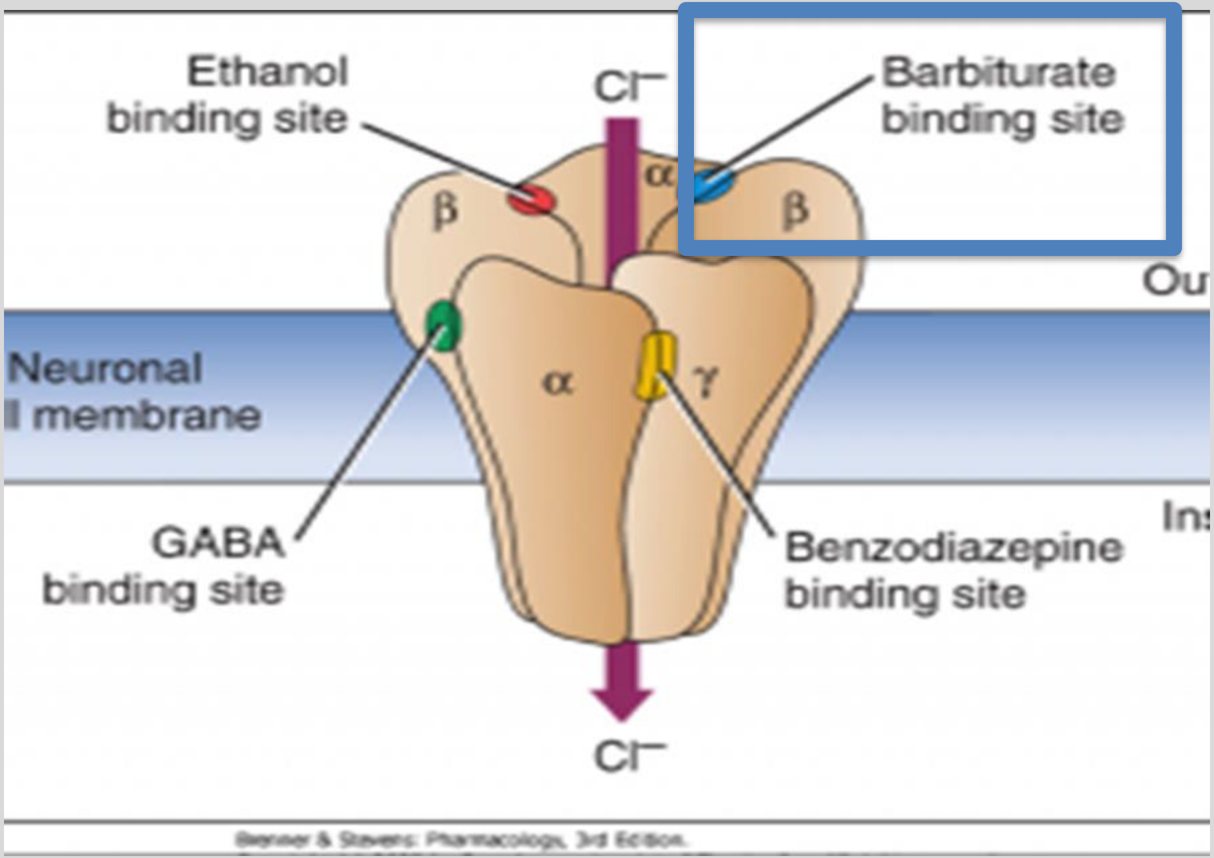
- Increase duration of GABA action. Opens the chloride gate in the absence of GABA.
- “No ceiling effect”
- Induce full respiratory arrest.
- Provide supportive care alkalinize urine

MOA: Potentiates GABA



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Barbiturate Adverse Effects

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- Highly sedative with a hangover effect
- Physical dependence: tolerance and withdrawal.
- Highly abused (pentobarbital)
- Also, meprobamate and chloral hydrate.

Benzodiazepines

- Chlordiazepoxide (Librium®) –long acting
- Clorazepate (Tranxene ®)- long acting
- Flurazepam (Dalmane ®)- long acting
- Diazepam (Valium ®)- long acting
- Lorazepam (Ativan ®)- intermediate acting
- Temazepam (Restoril®)- intermediate acting
- Alprazolam (Xanax®)-short acting
- Triazolam (Halcion®)- short acting
- Oxazepam (Serax®)- short acting
- Midazolam (Versed®)-short acting
- Flunitrazepam (Rohypnol)- “date-rape” drug

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Benzodiazepines

- Chlordiazepoxide and Diazepam (long acting). Alcohol detoxification.
- Alprazolam (short acting). Panic disorder.
- Lorazepam, Clonazepam, Diazepam (seizure disorder). 1st line for Status epilepticus
- Midazolam (anesthesia)
- Remimazolam (newest; rapid onset and offset)
- Estazolam and Flurazepam (insomnia)

Benzodiazepines

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- **Liver metabolism (converted to active metabolites).**
- **Chlordiazepoxide, diazepam, flurazepam (metabolites make the drug long acting). Appropriate for detoxification protocols.**
- **Oxazepam, temazepam, and lorazepam are safest for elderly. Bypass phase I metabolism (only conjugated).**
- **<u>Use all benzos cautiously in elderly.</u>**

Benzodiazepine Biotransformation

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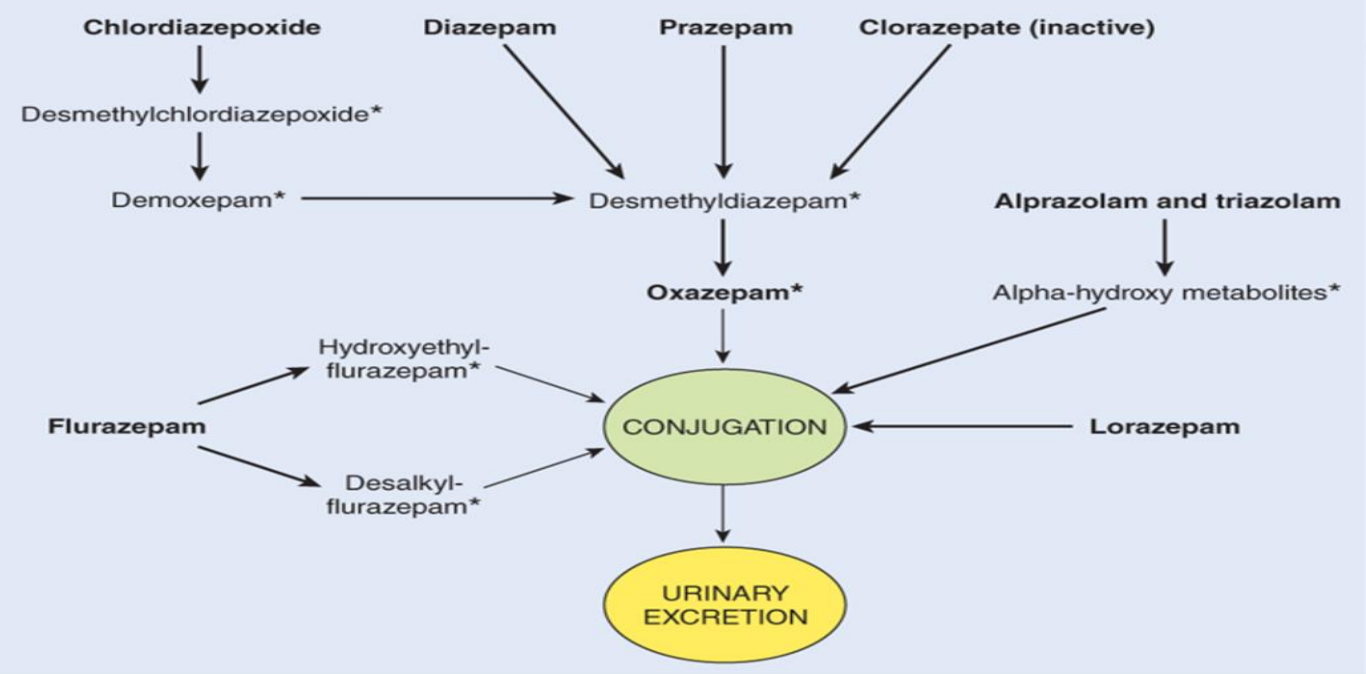

Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition

Benzodiazepines

- Potentiates GABA by increasing frequency of chloride channel opening. Needs GABA to work.
- "Ceiling effect"
- No respiratory depression, unless combined with other CNS depressants.
- Use: anxiety, sedative-hypnotic, insomnia anterograde amnesia, muscle relaxant, anticonvulsant, anticipatory N/V

MOA: Potentiates GABA

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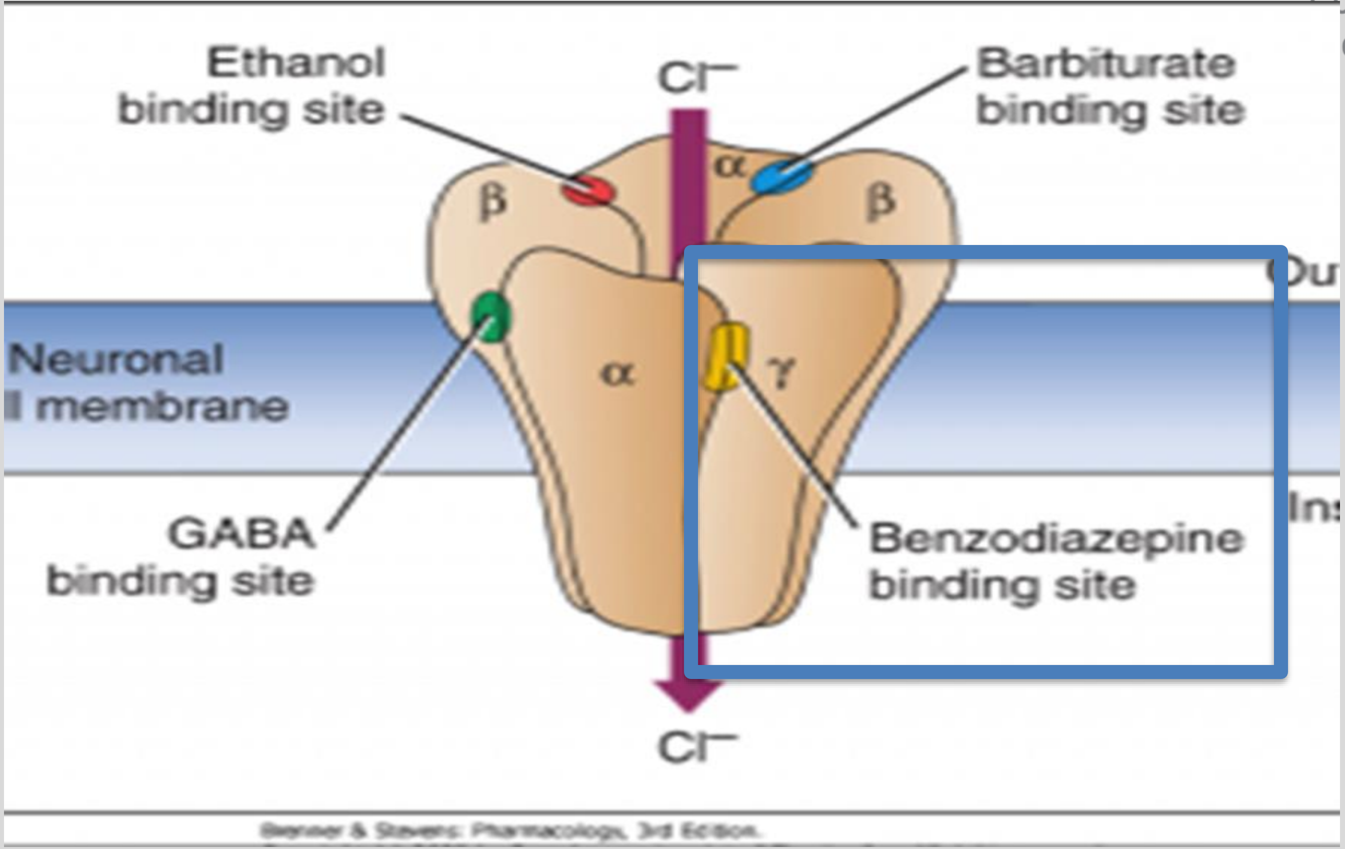
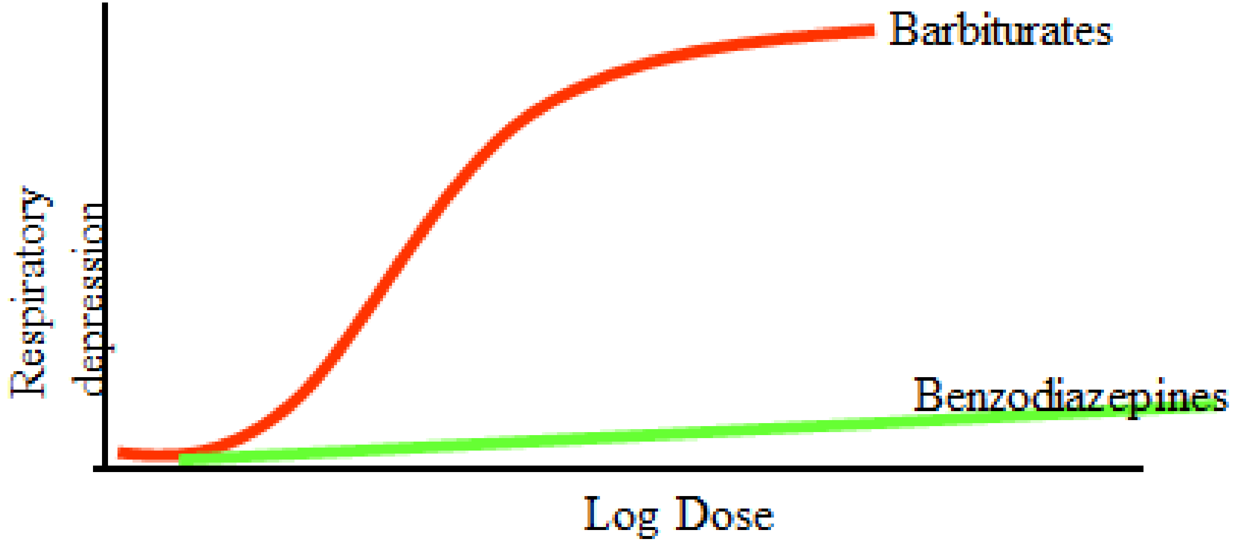


Benzos vs Barbituartes



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Benzodiazepines Adverse Effects

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- Drowsiness, hangover
- Mild euphoria (abuse potential)
- Anterograde amnesia
- Physical dependence: tolerance and withdrawal symptoms: rebound anxiety, insomnia, seizures (alprazolam).
- Fetal malformations “floppy infant”

Flumazenil

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- Benzodiazepine antagonist, IV
- Reverses benzodiazepine toxicity in overdose.
- When flumazenil is used after midazolam, this drug will rapidly result in patient recovery.
- Caution in chronic BDZ users, precipitates "withdrawal"

Non-Benzodiazepines “z drugs”

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- Zolpidem, zaleplon, eszopiclone
- Minimal effect on REM sleep (unlike barbs and BDZ).
- Shorter duration of action
- Less hangover effect
- Target specific $GABA_A$ subunit.
- Potentiated by ethanol

Ramelteon, Tasimelteon, and Melatonin

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- MT-1and MT-2 melatonin agonists. No effect on GABAergic neurotransmission. Not a controlled substance. Substrate for CYP $1A_2$.
- Melatonin: released from pineal gland, produces drowsiness. Used for jet lag.

Other “Sedative” Drugs

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- Suvorexant, lemborexant, and daridorexant (orexin antagonists at OX1, OX2 receptors)
- H_1 receptor antagonists (1st Generation only) Adverse effects?
- Tricyclic antidepressants (doxepin)
- Trazodone (H_1 blockade)
- Ethanol (avoid with other CNS “depressants”)

H_1 receptor mechanism (Gq)

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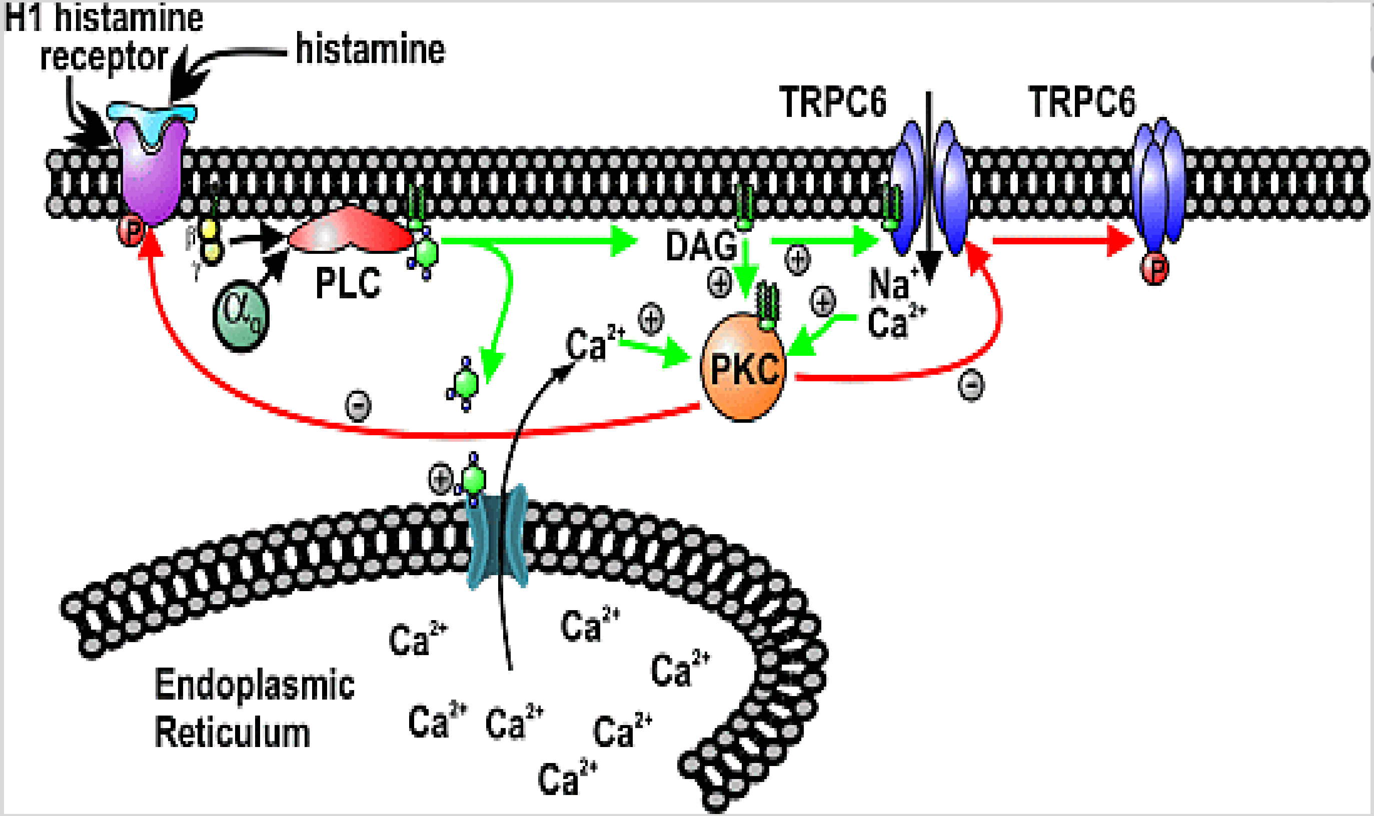

Tricyclic Antidepressants

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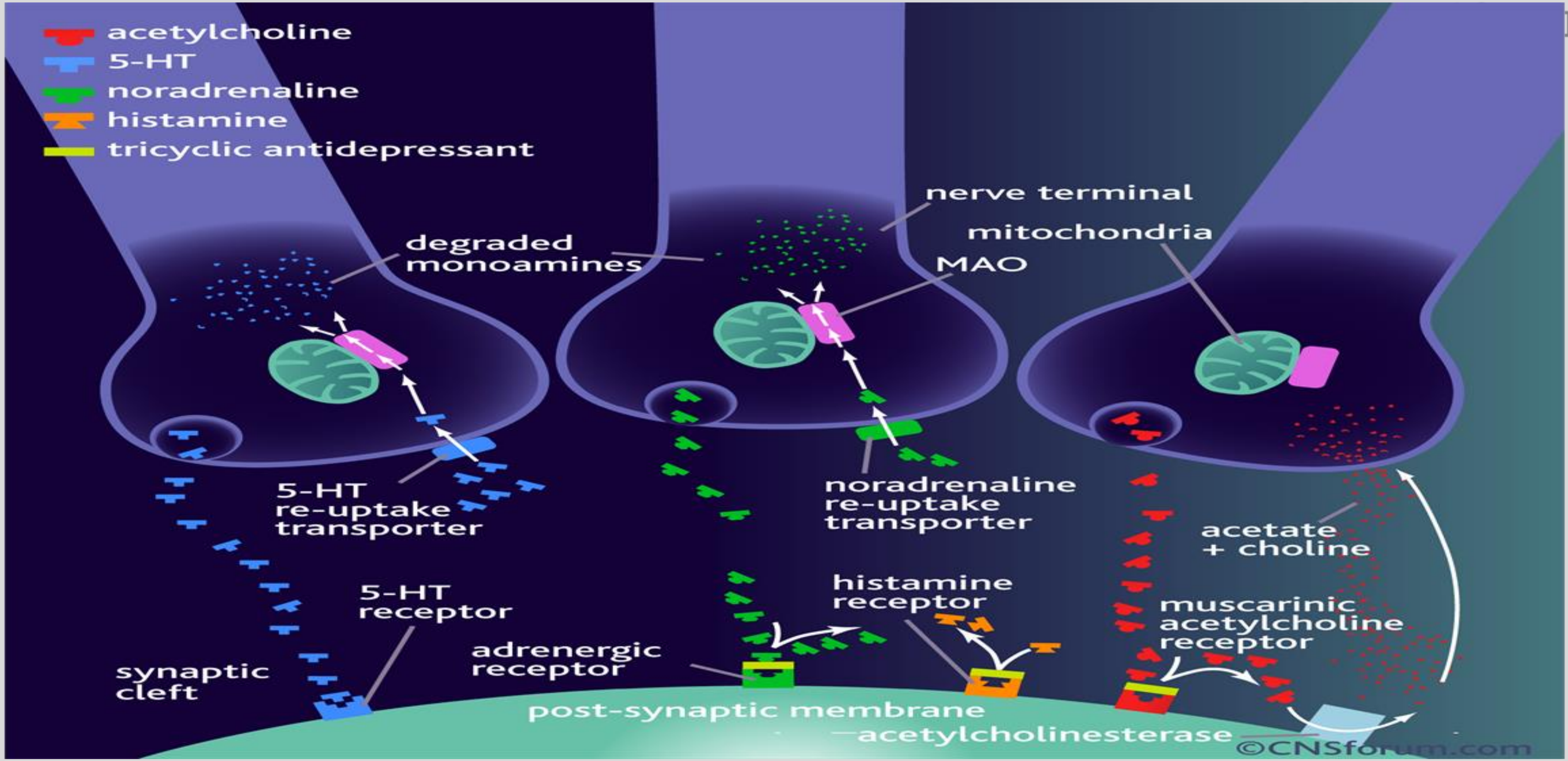

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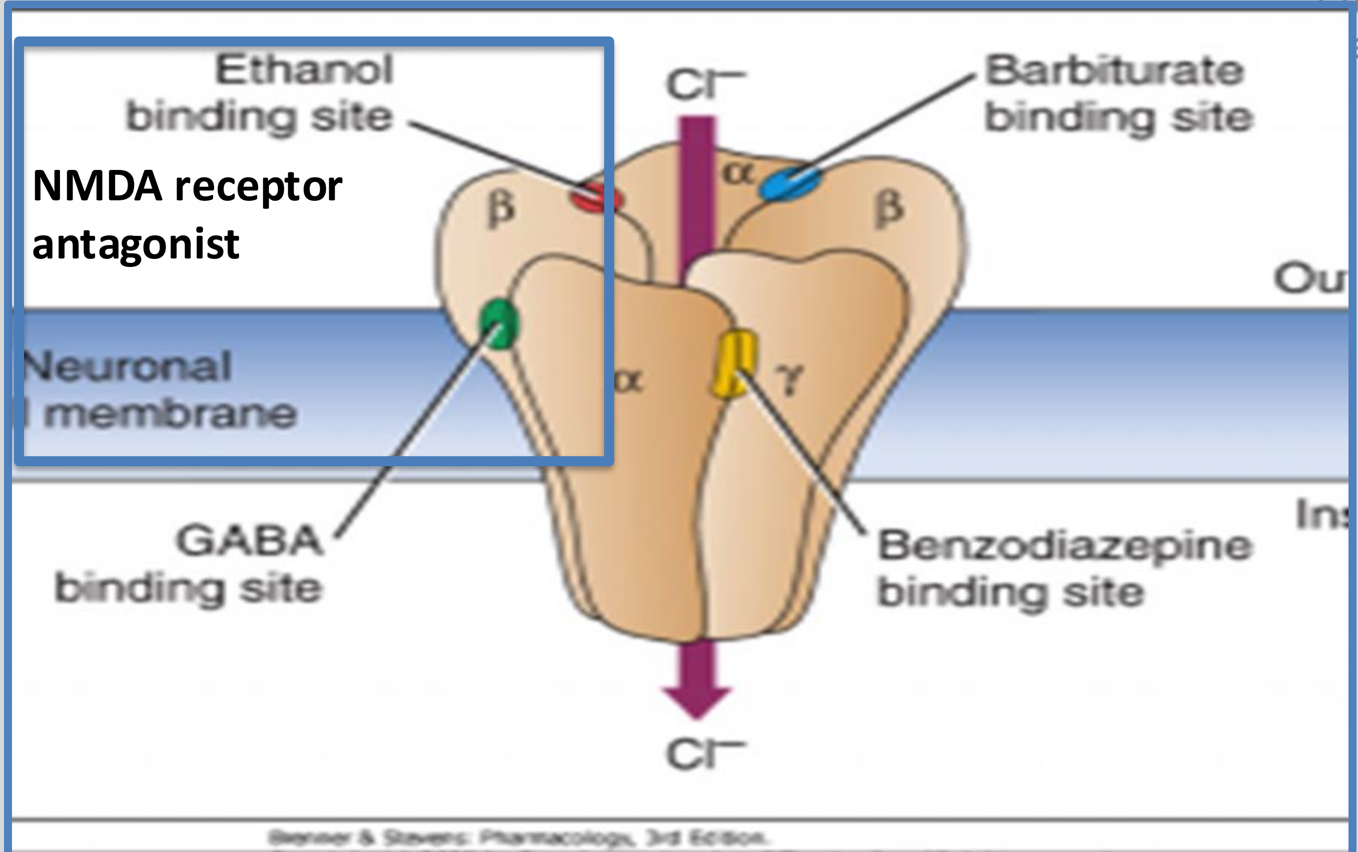
Ethanol: Potentiates GABA



Ethanol (chronic)

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- Dilated cardiomyopathy
- Thiamine deficiency (Wernicke's encephalopathy): thiamine before glucose.
- Fetal alcohol syndrome
- Zero order kinetics
- Induces CYP 2E1 (enhanced acetaminophen toxicity)
- Increased inflammation/neoplasia
- Loss of liver function

Ethanol: Oxidative Metabolism

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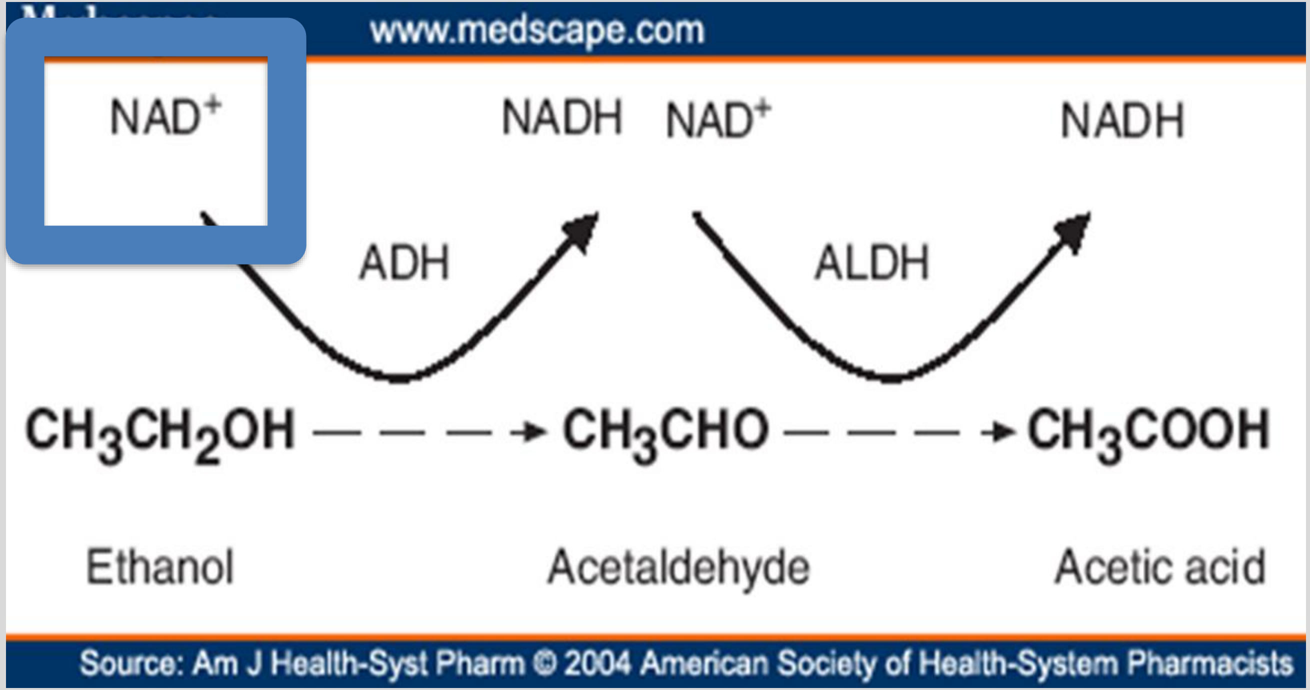

High NADH:NAD ratio with ethanol consumption

1. **Lactate accumulates**
2. **Increases fatty acid**
3. **Lose glutathione**

Fetal Alcohol Syndrome

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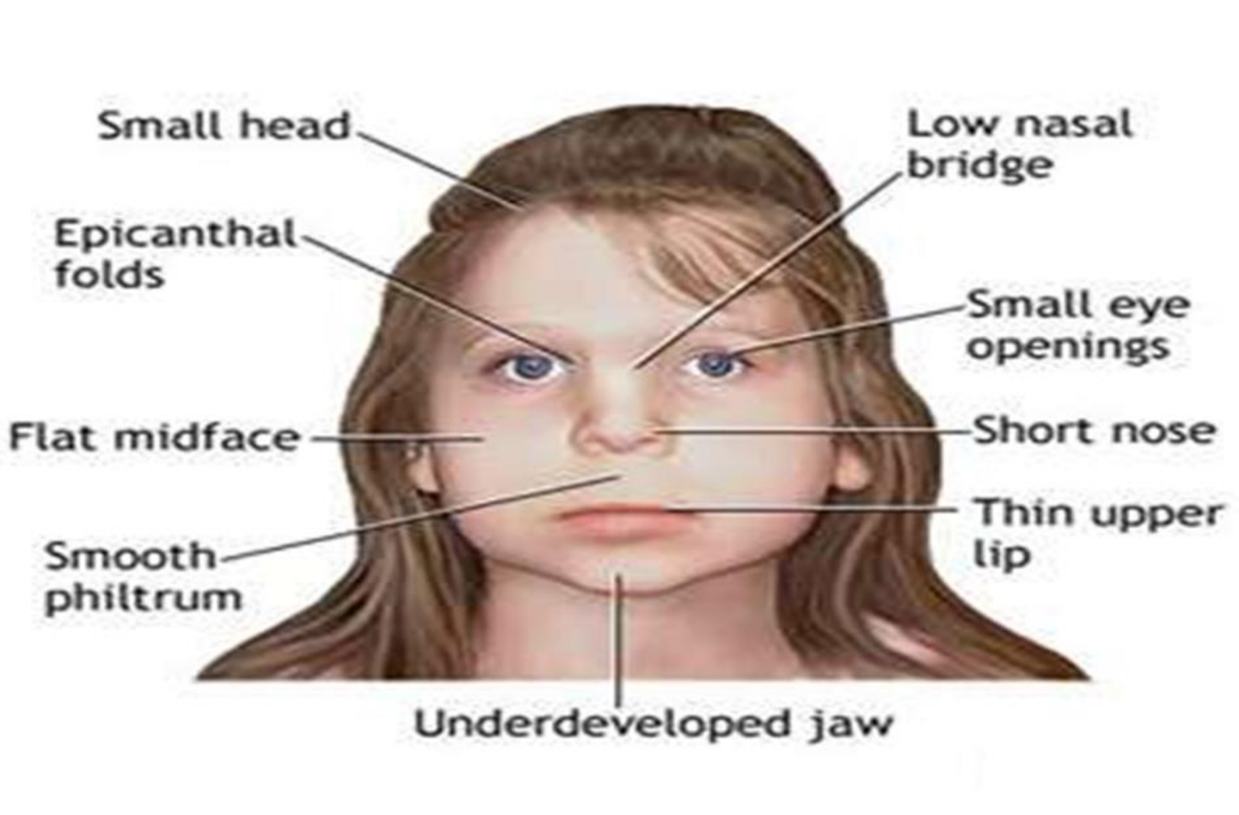
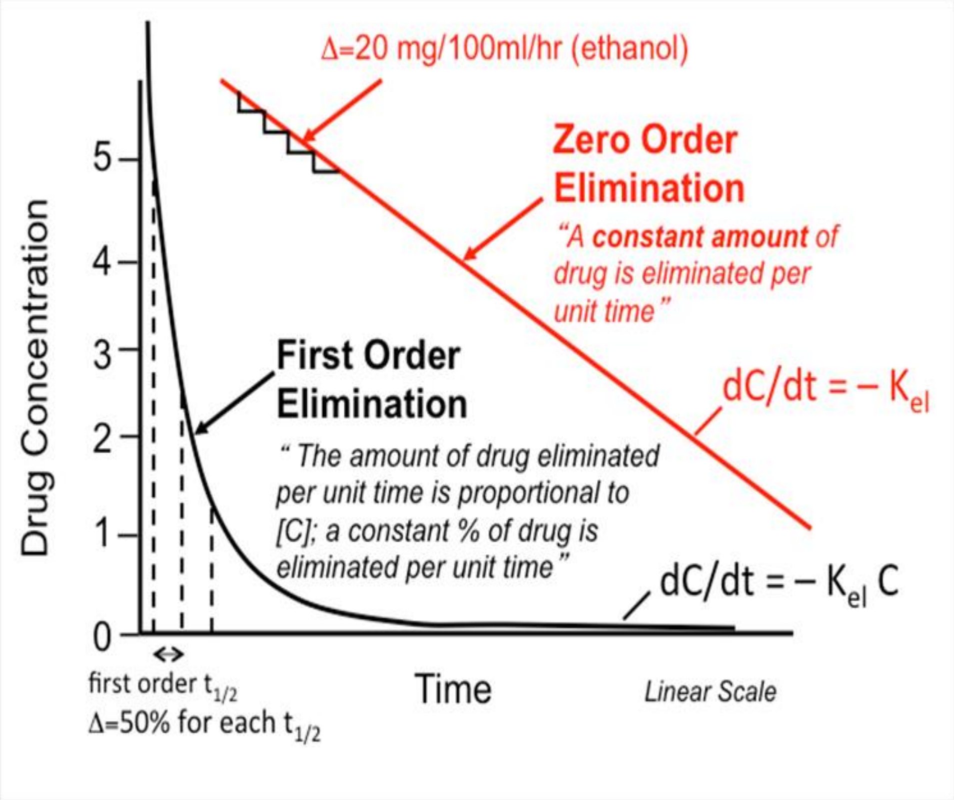


Zero Order Elimination



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Ethanol Intoxication

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- Slurred speech, nystagmus, stupor, or coma.
- Hypotension and tachycardia
- Hypoglycemia, lactic acidosis, hypokalemia, hypomagnesemia
- Dehydration
- Treatment: supportive care, ventilation
- 80mg/dL BAC (intoxication)

Ethanol Withdrawal

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Syndrome	Clinical findings	Onset after last drink
Minor withdrawal	Tremulousness, mild anxiety, headache, diaphoresis, palpitations, anorexia, GI upset; Normal mental status	6 to 36 hours
Seizures	Single or brief flurry of generalized, tonic-clonic seizures, short post-ictal period; Status epilepticus rare	6 to 48 hours
Alcoholic hallucinosis	Visual, auditory, and/or tactile hallucinations with intact orientation and normal vital signs	12 to 48 hours
Delirium tremens	Delirium, agitation, tachycardia, hypertension, fever, diaphoresis	48 to 96 hours

Up-to-date, 2016

Ethanol Overdose, Withdrawal, and Alcohol Use Disorder

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- Overdose: maintain airway, supportive care
- Symptoms of withdrawal: CNS stimulation, seizures, "delirium tremens" BENZO must be administered. Long-acting Benzo (ex. Chlordiazepoxide).
- Alcohol use disorder: Naltrexone (opioid blocker); Acamprosate (restores GABA/glutamate balance); CBT

Disulfiram: Antabuse

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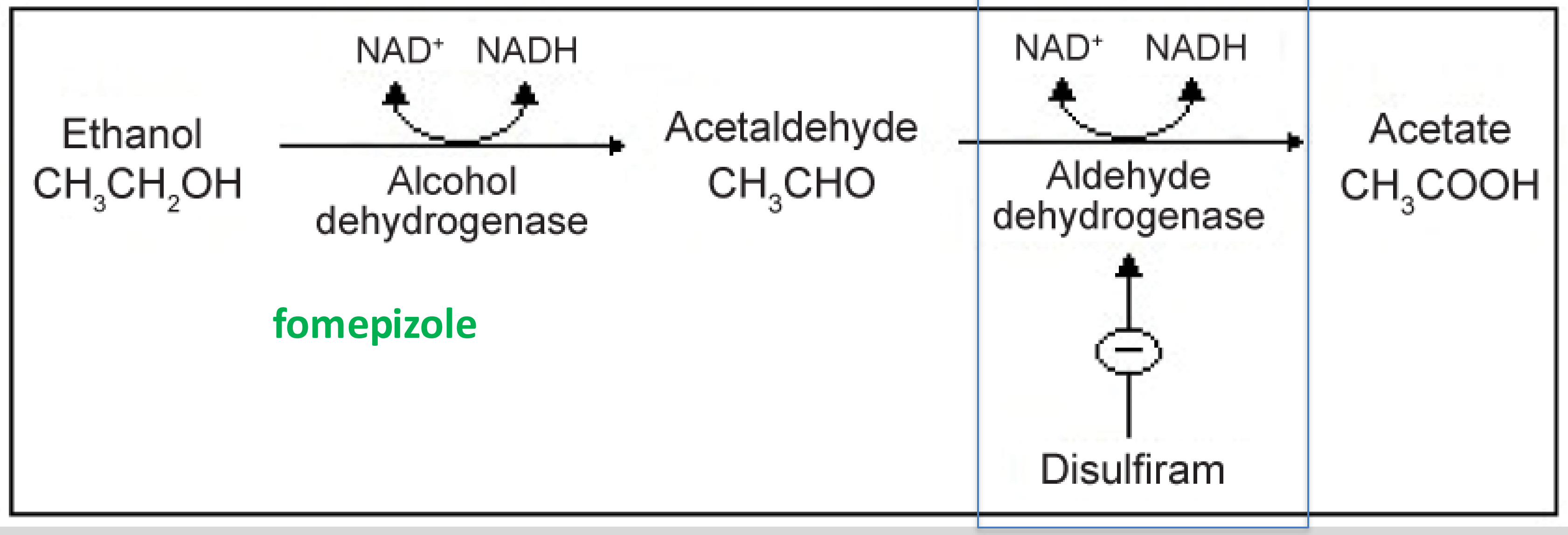

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Antabuse
Metronidazole
Chlorpropamide

Clonidine

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- Alpha$_2$ adrenergic agonist
- Reduce sympathetic NS effects following alcohol, opioid, and nicotine withdrawal.
- Adverse effects: sedation, depression, bradycardia, hypotension

Anxiety

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- Natural pattern of behavior
- Heightened state of arousal and preparedness
- Elevated sympathetic N.S.
- Central Nucleus of Amygdala has a variety of neurotransmitters: GABA, Glu, 5-HT, NE
- Benzos constitute the most used group of anxiolytics.

Anxiety Classification and Treatment

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- Acute anxiety: Benzodiazepine
- Panic disorder: Alprazolam, then SSRI for long-term.
- Phobic disorder: Benzo then SSRI.
- Propranolol ("Situational anxiety")
- OCD: SSRI and psychotherapy
- Generalized anxiety disorder (GAD): Benzo, then SSRI, SNRI, or buspirone

Propranolol

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- Non-selective β adrenergic blocker
- Stage Fright “situational anxiety”
- Anxiety involves the Sympathetic N.S.
- B_2 activity: caution in asthma, diabetes, peripheral vascular disease

Buspirone

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- Partial $5HT_{1A}$ agonist, not GABA.
- Weak D_2 antagonist
- Use: chronic anxiety (GAD)
- Reduces anxiety in absence of tolerance, rebound, or dependence.
- Full anxiolytic effect may be 4-6 weeks
- Adverse effects: dizziness

Narcolepsy

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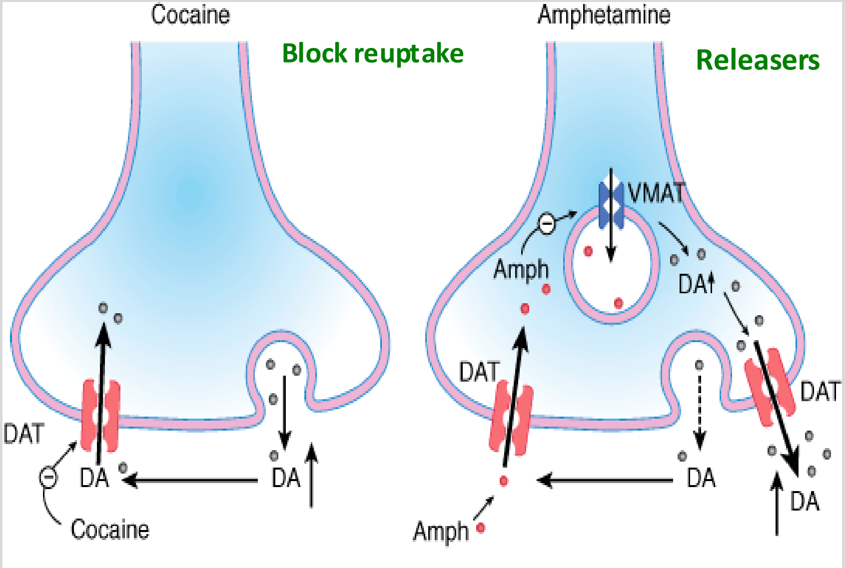
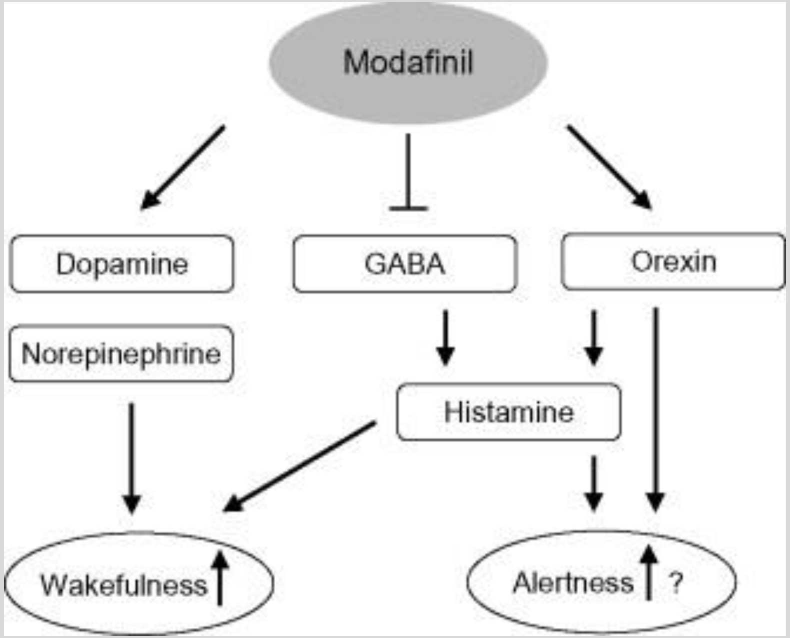
- Daytime sleepiness, despite sleeping at night
- Treat with sympathomimetics “stimulants”, also for ADHD
- Amphetamine, dextroamphetamine, and methylphenidate inhibit reuptake of NE and DA. Also increases release of catecholamines from nerve terminals.
- Adverse effects: Cardiovascular, weight loss, anxiety.
- Modafinil and armodafinil (greater DA vs NE reuptake inhibition and release). Less anxiety and anorexia than amphetamines. Skin rash reported. Activates orexin, inhibits GABA.

Cocaine vs Amphetamine



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 11th Edition*: http://www.accessmedicine.com

Modafinil



Stimulant Overdose

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- Symptoms of overdose: difficult to differentiate schizophrenia symptoms, tachycardia, mydriasis, hypertension, hyperthermia, seizures.
- "Cocaine bugs"
- Overdose: acidify urine, supportive therapy (lorazepam, haloperidol, and vasodilator). Labetalol

Sodium Oxybate

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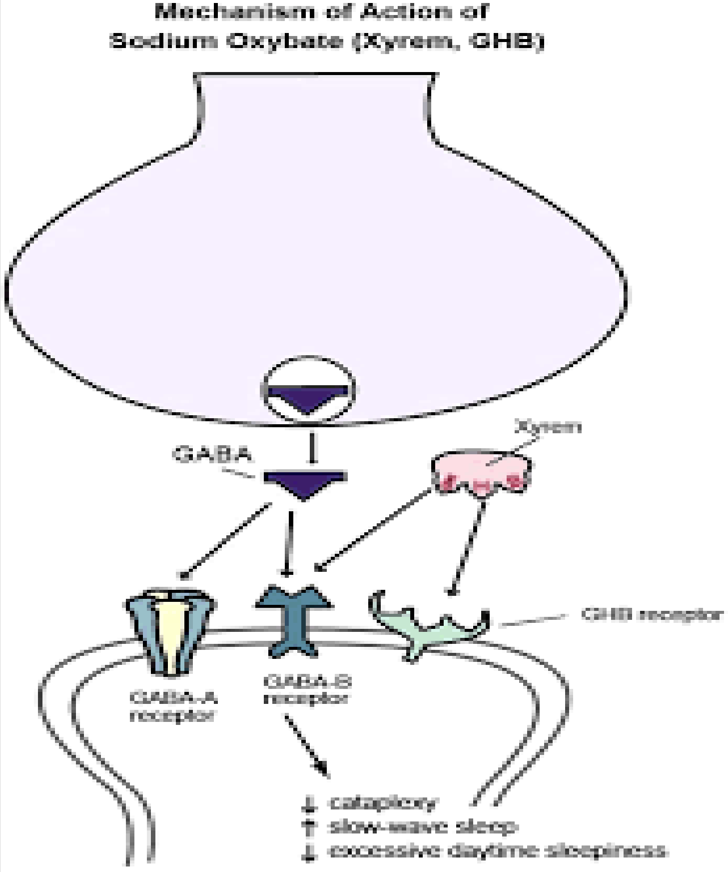

- Treats narcolepsy with cataplexy.
- Sodium salt of gamma-hydroxybutyrate (GHB), metabolite of GABA.
- Agonist of $GABA_B$ receptor, short $t_{1/2}$
- Risk Evaluation and Mitigation (REMS) program. Mitigates the risk of abuse and misuse.
- AE: N/V, confusion, not to be administered with ethanol or sedative hypnotics (increased risk of respiratory depression).
- Stimulant may be taken during day

Summary Slide

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- Describe the differences between barbs, benzos, and ethanol in MOA and safety.
- Describe drugs used to induce sleep and manage narcolepsy

Core References:

1. Basic & Clinical Pharmacology, 16e. Bertram G. Katzung, Todd Vanderah; Chapter 22, Sedative-Hypnotic Drugs; Chapter 23, The Alcohols; Chapter 32, Drugs of Abuse
2. Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e, Chapter 22, Hypnotics and Sedatives; Chapter 27, Ethanol
3. Sedative-Hypnotics Summary Guide and Practice Questions (Maria A. Pino, Ph.D)

Lecture Feedback Form:

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